Kairav Session 30

Sunday, October 18, 2026 8:35 PM

Counting Sort

list = [5, 4, 4, 3, 1, 7, 7, 2, 5, 8, 7, 6, 5, 3, 2, 2]

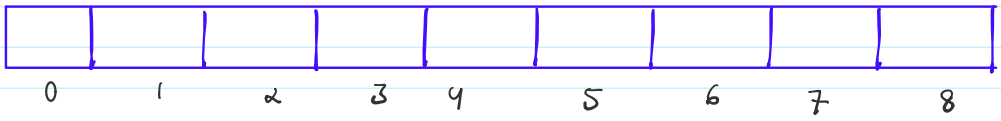
Note :- List elements should be positive (list[i] ≥ 0)

Step 1: Find the maximum element of the list

Max = 8

step 2: Create empty frequency of size max + 1 → 9

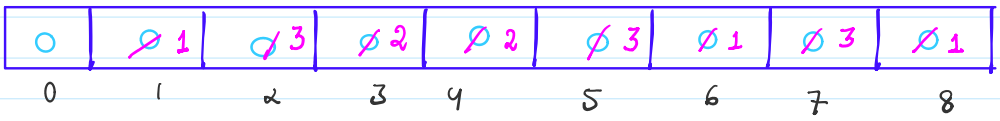
freq →



step 3: Store the frequency of every elements in the frequency list

list = [5, 4, 4, 3, 1, 7, 7, 2, 5, 8, 7, 6, 5, 3, 2, 2]

freq →

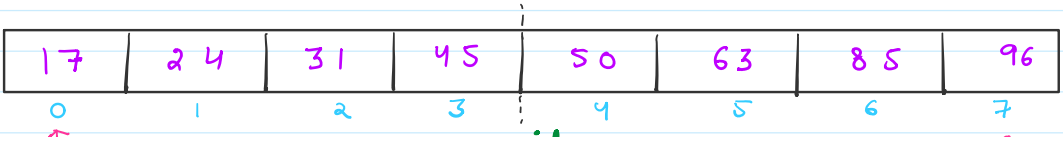


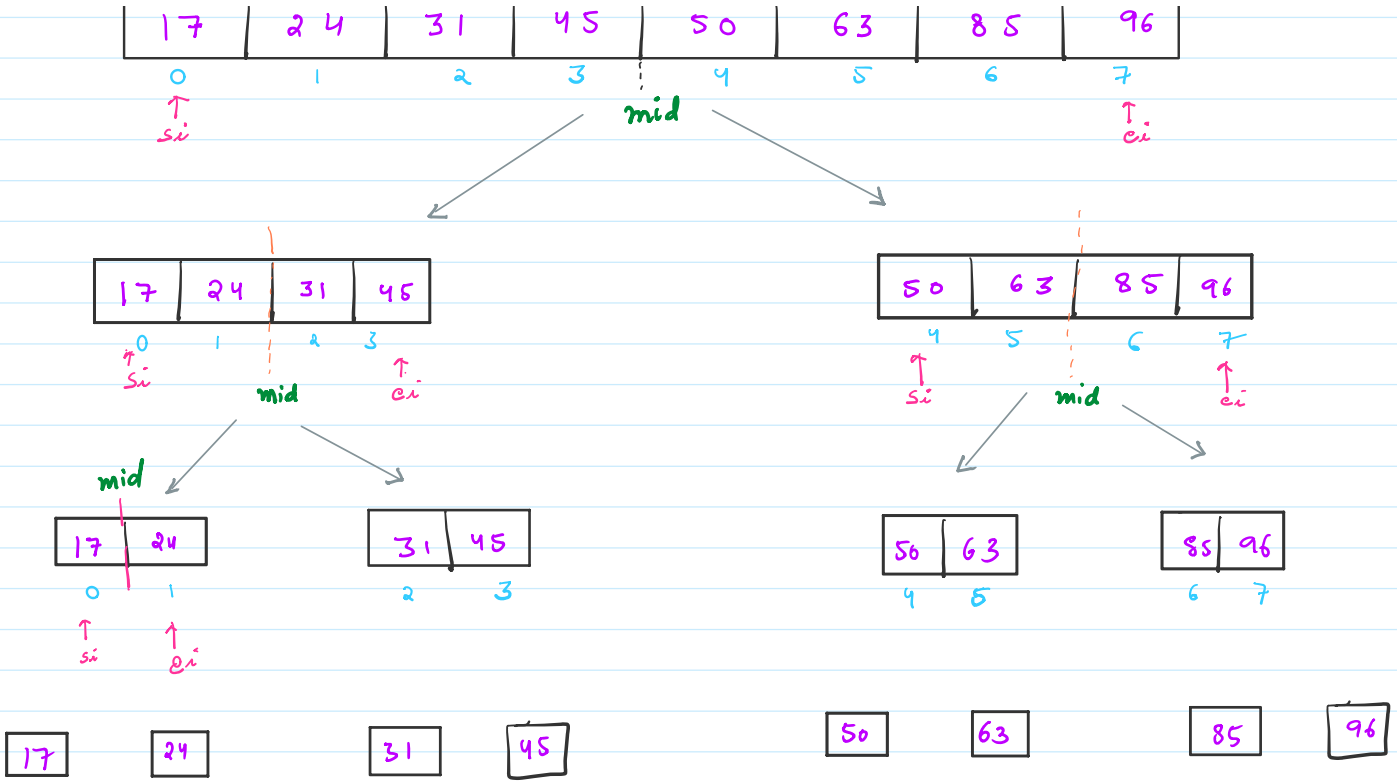
step 4:- 1 2 2 2 3 3 4 4 5 5 5 6 7 7 7 8

Merge Sort :

→ It is a divide & conquer sorting algorithm.

Prerequisite:- Recursion




17 24 31 45 50 63 85 96
0 1 2 3 4 5 6 7
si
mid
ei
17 24 31 45
0 1 2 3
si
mid
ei
50 63 85 96
4 5 6 7
si
mid
ei
mid
17 24
0 1
si
ei
31 45
2 3
50 63
4 5
85 96
6 7
17
24
31
45
50
63
85
96